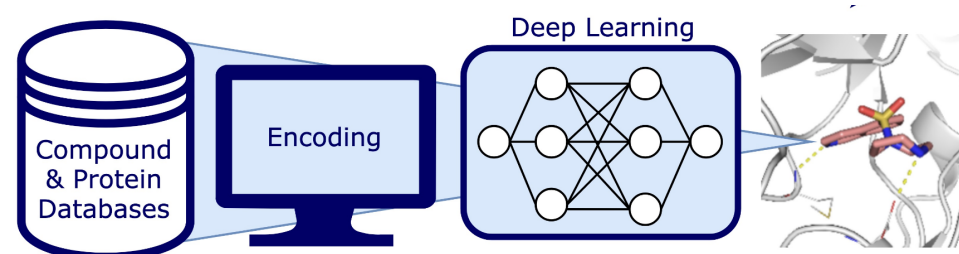


Data Driven Drug Design

Machine Learning with Python

Prof. Dr. Andrea Volkamer
Data Driven Drug Design Group



What we will do in detail ...

Session 1

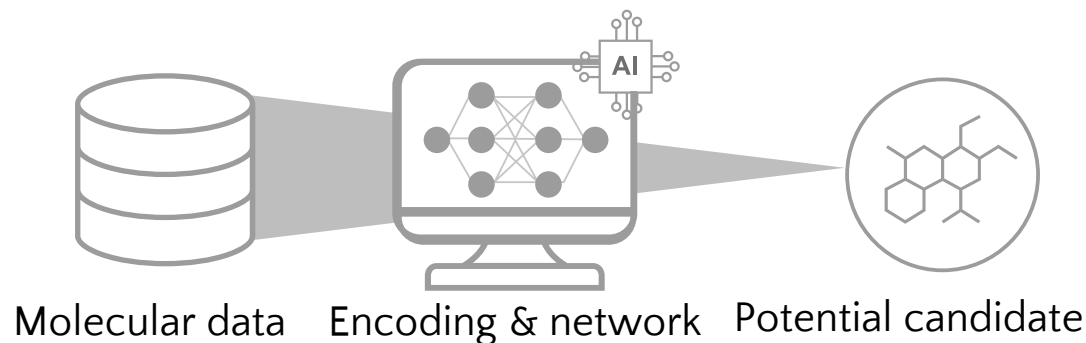
- CADD, Artificial Intelligence (AI) and Machine Learning (ML)
- The importance of data quality for AI/ML: Data preprocessing and analysis
- Molecular representations: Fingerprints and similarities

Session 2

- **Introduction machine learning (ML)**
- **(Un)supervised learning: RF, SVM, ...**
- Performance evaluation

Session 3

- Deep learning
- Neuronal networks
- Molecular encodings and architectures



Artificial Intelligence, Machine & Deep Learning

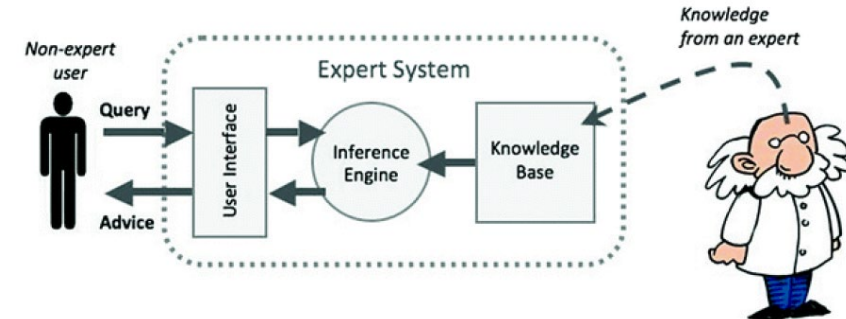
Artificial Intelligence (AI)

Machine Learning (ML)

Deep Learning (DL)

AI: Rule-based learning

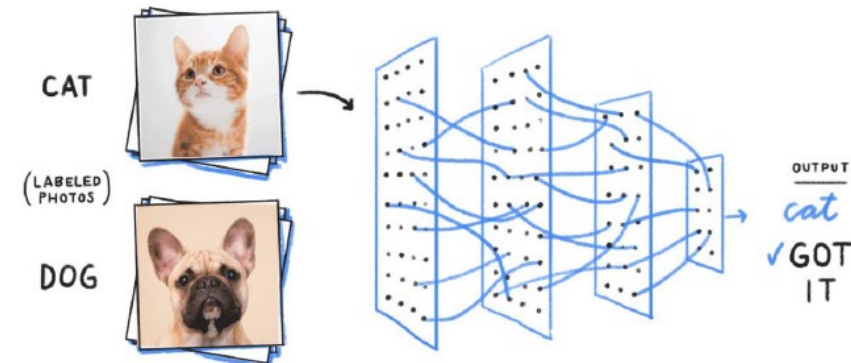
- Development of computer systems
- Performing tasks requiring human intelligence



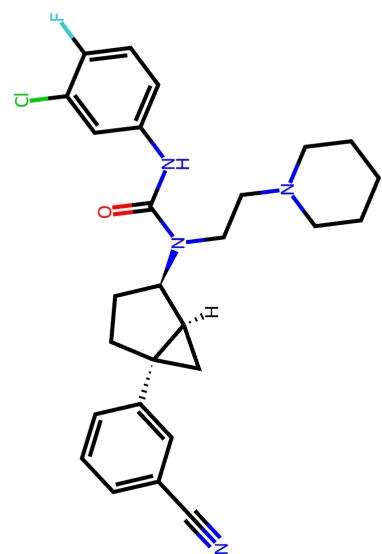
https://link.springer.com/chapter/10.1007/978-981-10-9035-6_82

ML: Learning from examples, human-made descriptors

DL: Deep neural networks, learning representations



The Path From Molecules to Property Predictions



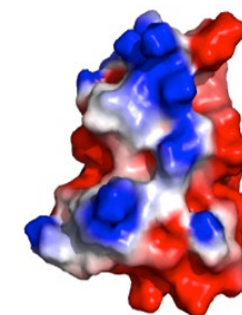
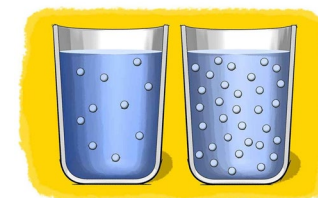
Supervised Learning (SL)

Representation

- Physchem properties
- Morgan FPs
- Adjacency matrix

Model

- RF
- SVM
- MLP/NN
- GNN



Self-Supervised Learning (SSL)

Tokenization

- Sequences
- Graphs

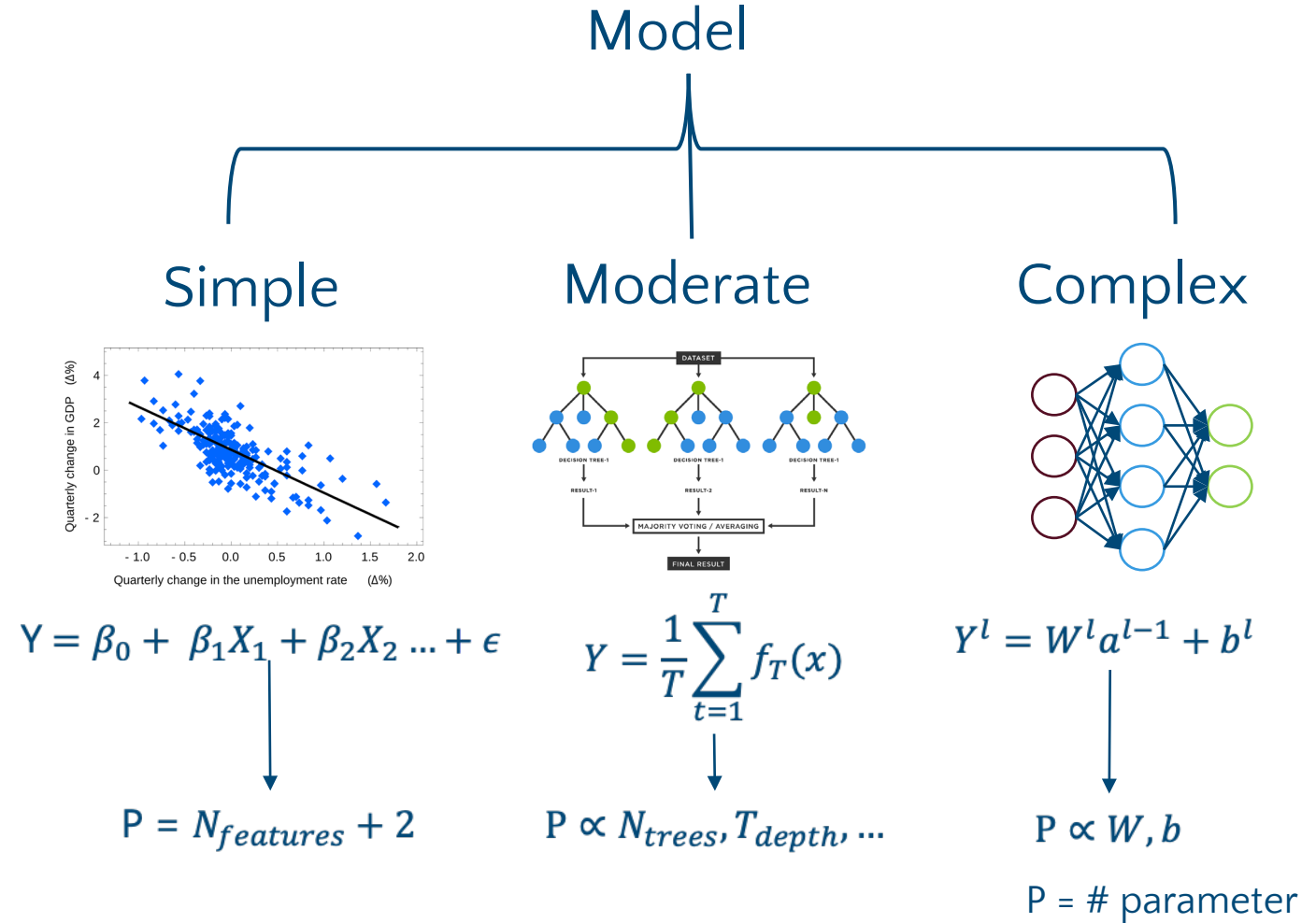
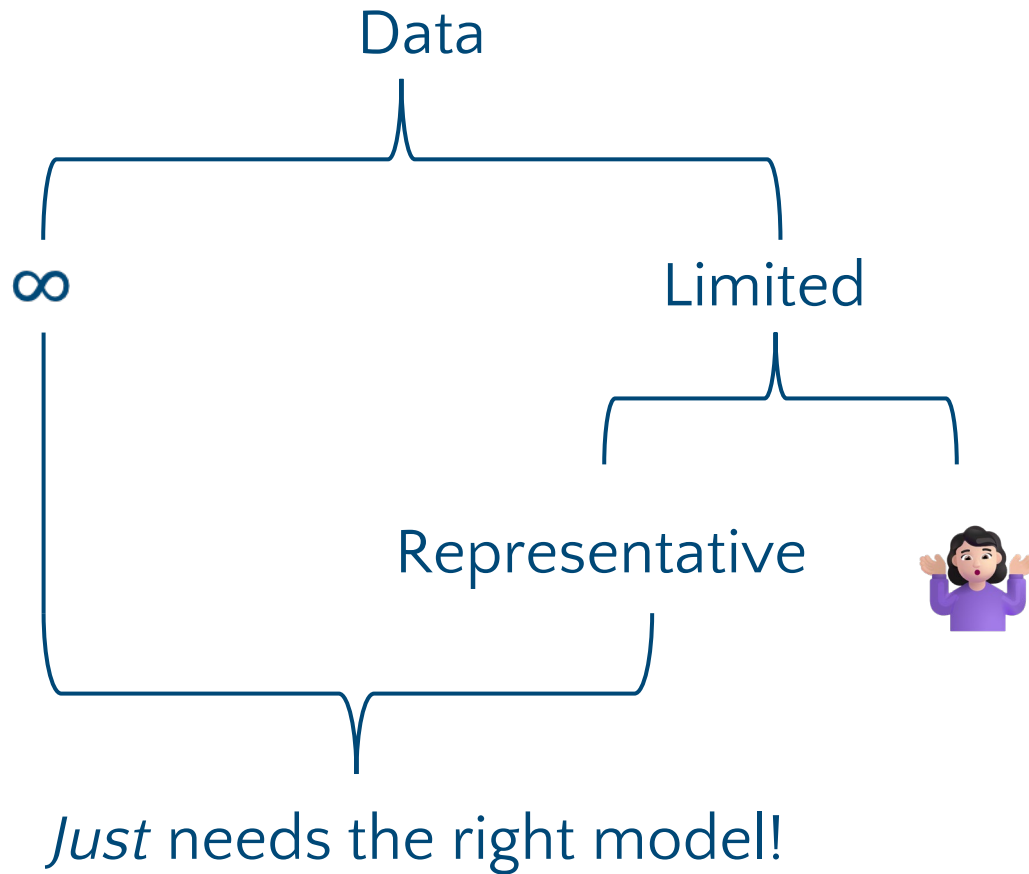
Foundational Models

- Transformers
- Autoencoders

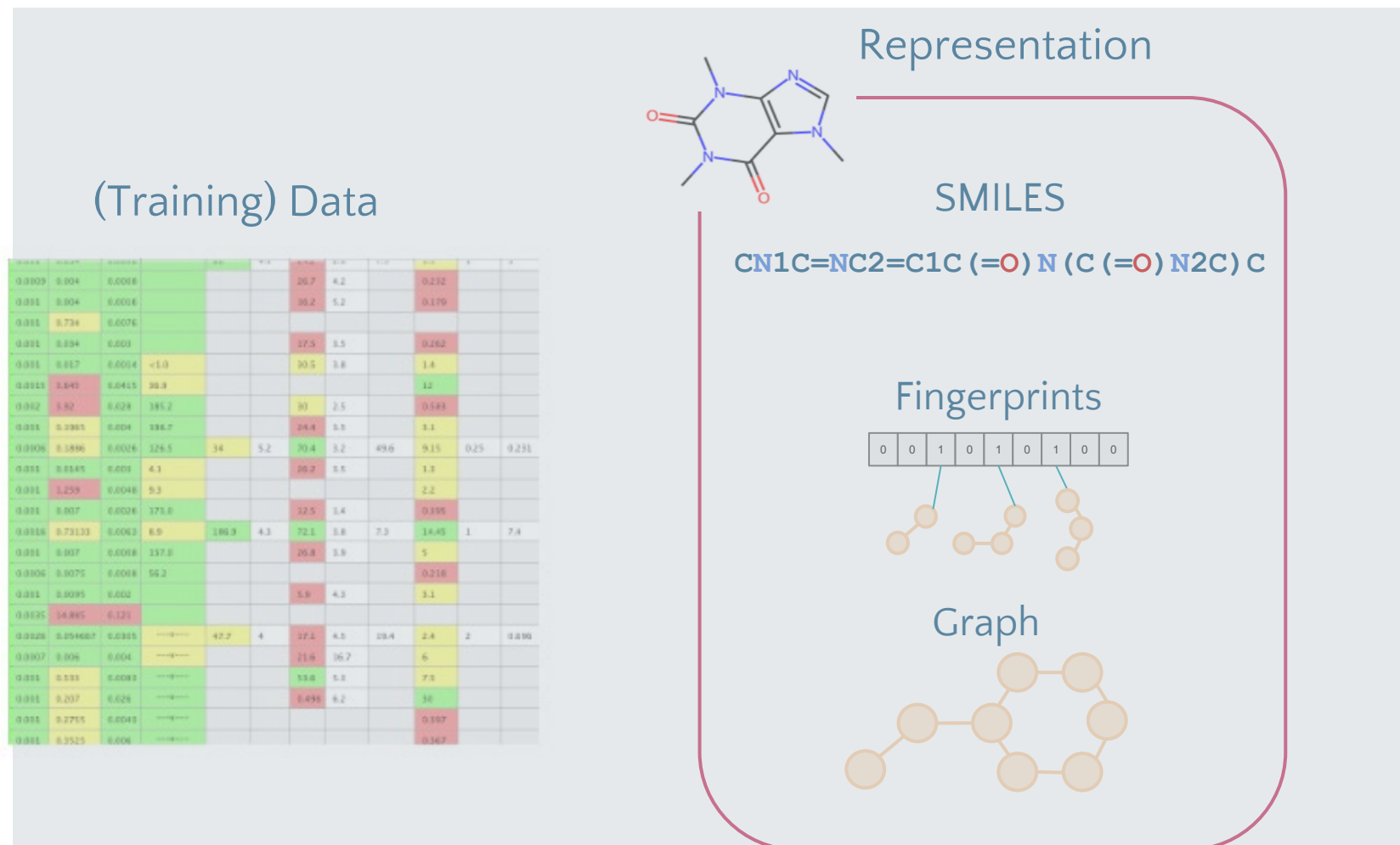
Fine-tuning



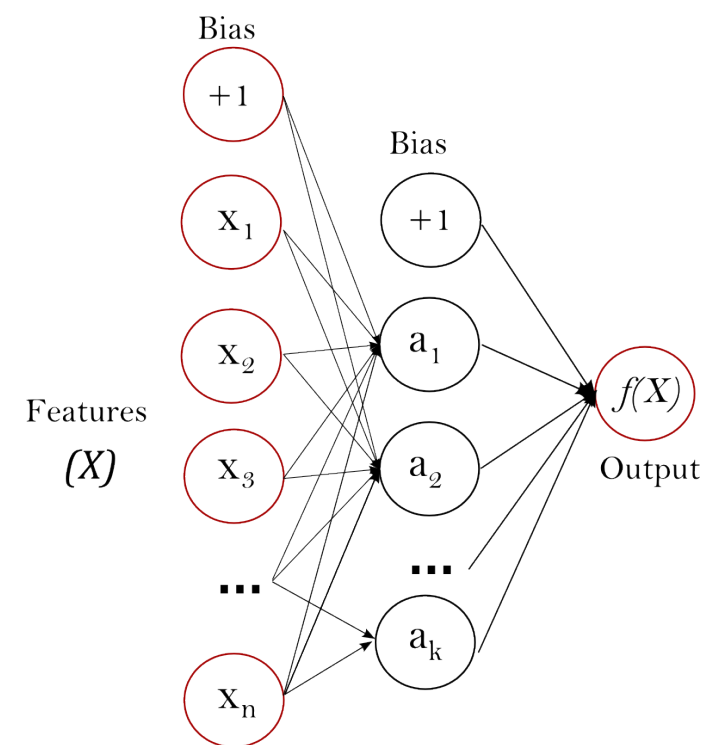
The Two Limiting Factors



Key Components of Machine Learning in Drug Discovery (or Anything Else)

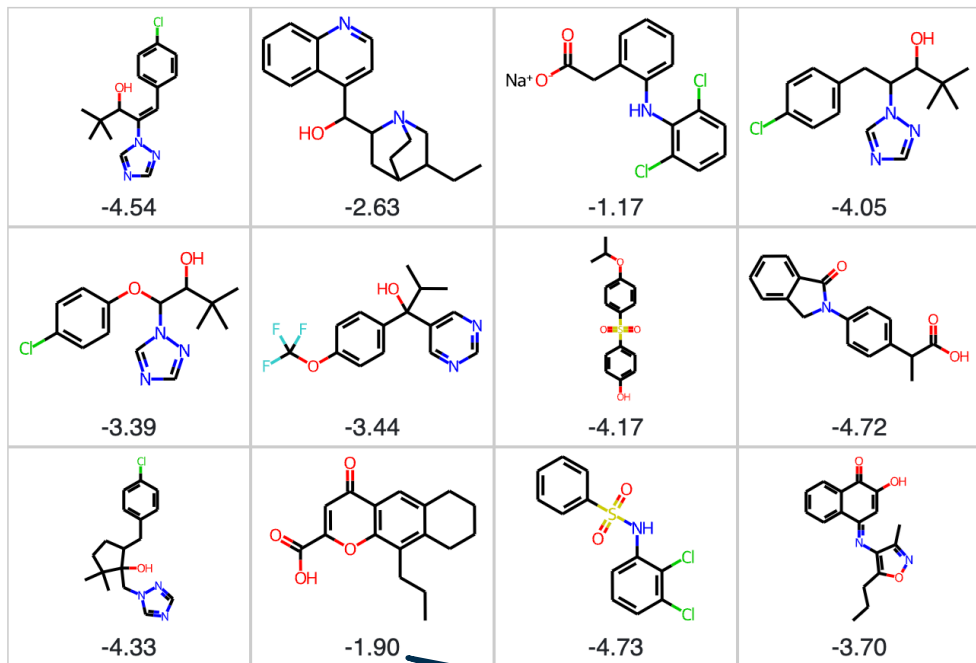


Architecture

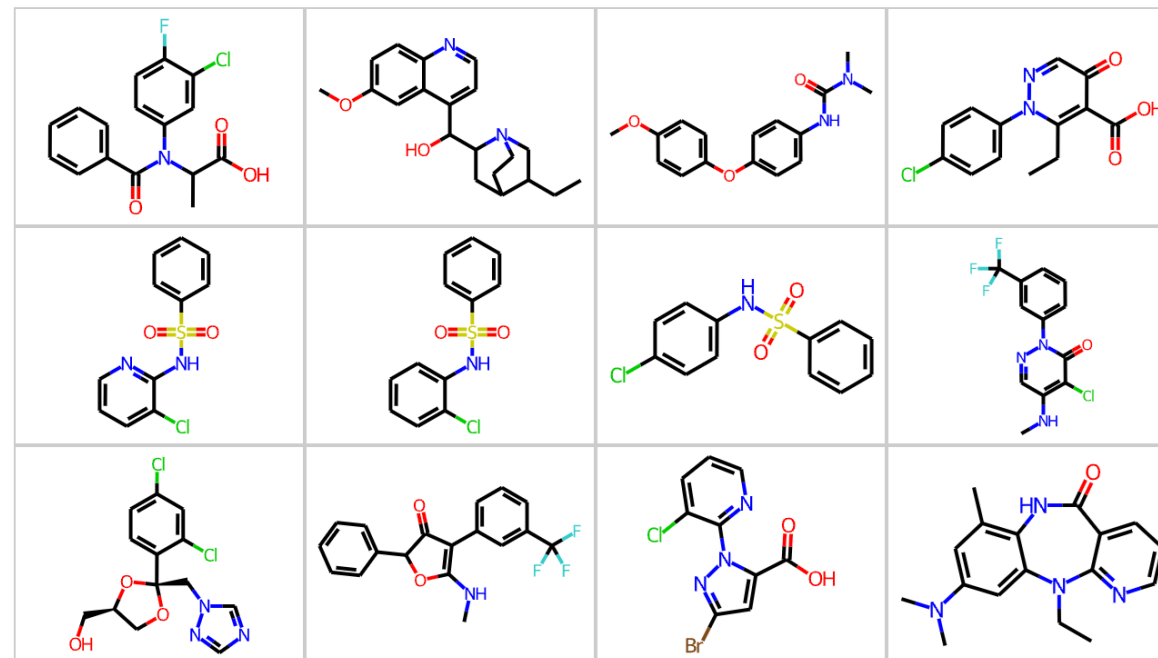


Modern QSAR/ML/AI: Labeling Molecules Based on Examples

Molecules with measured data

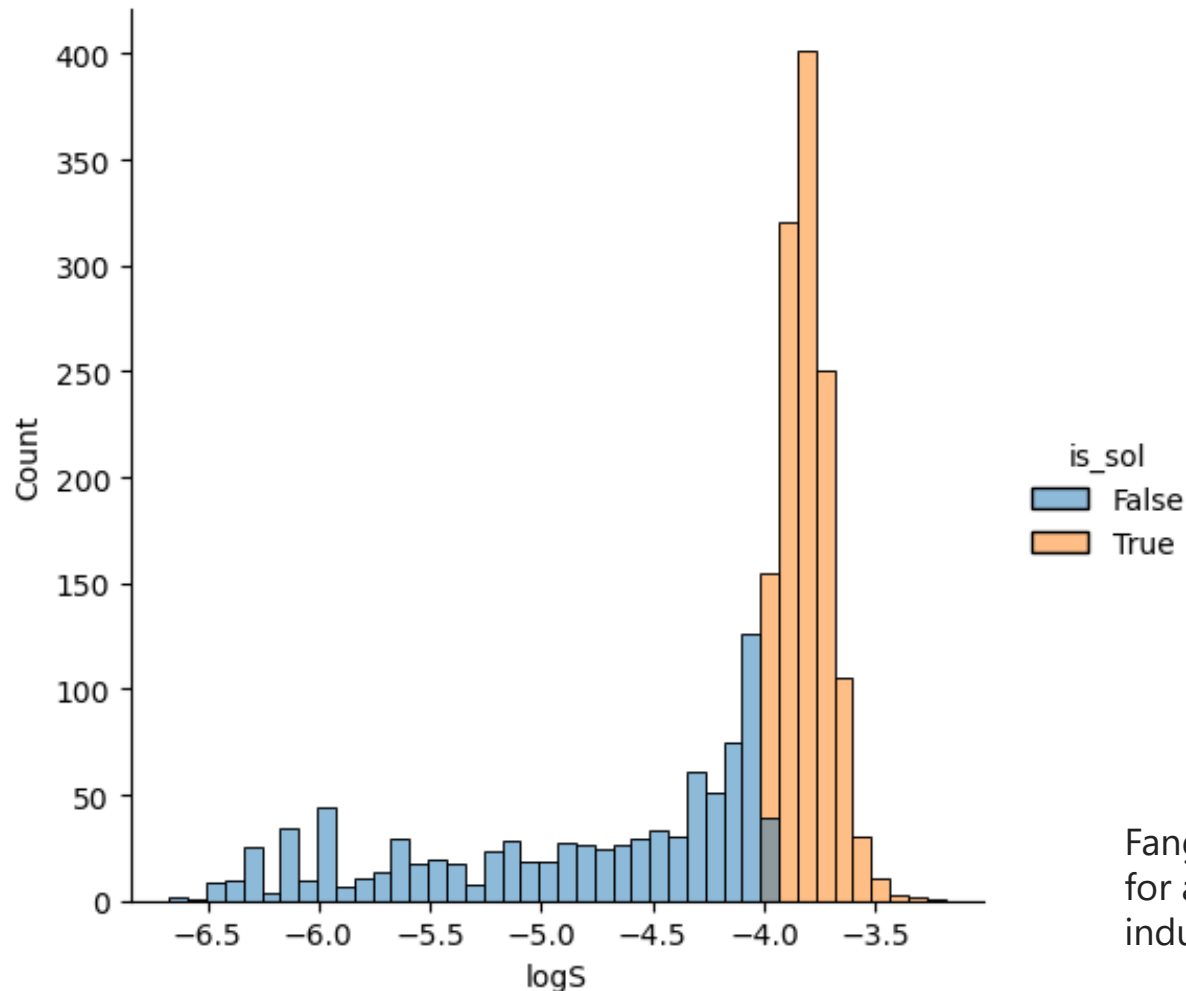


Molecules to be predicted



$\text{Log}_{10}(\text{Molar Aqueous Solubility})$

Let's Start With a Dataset from *Polaris*

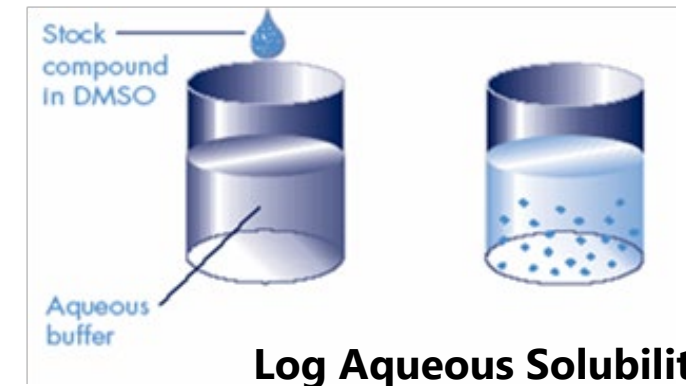


Aqueous solubility by CLND

2173 compounds

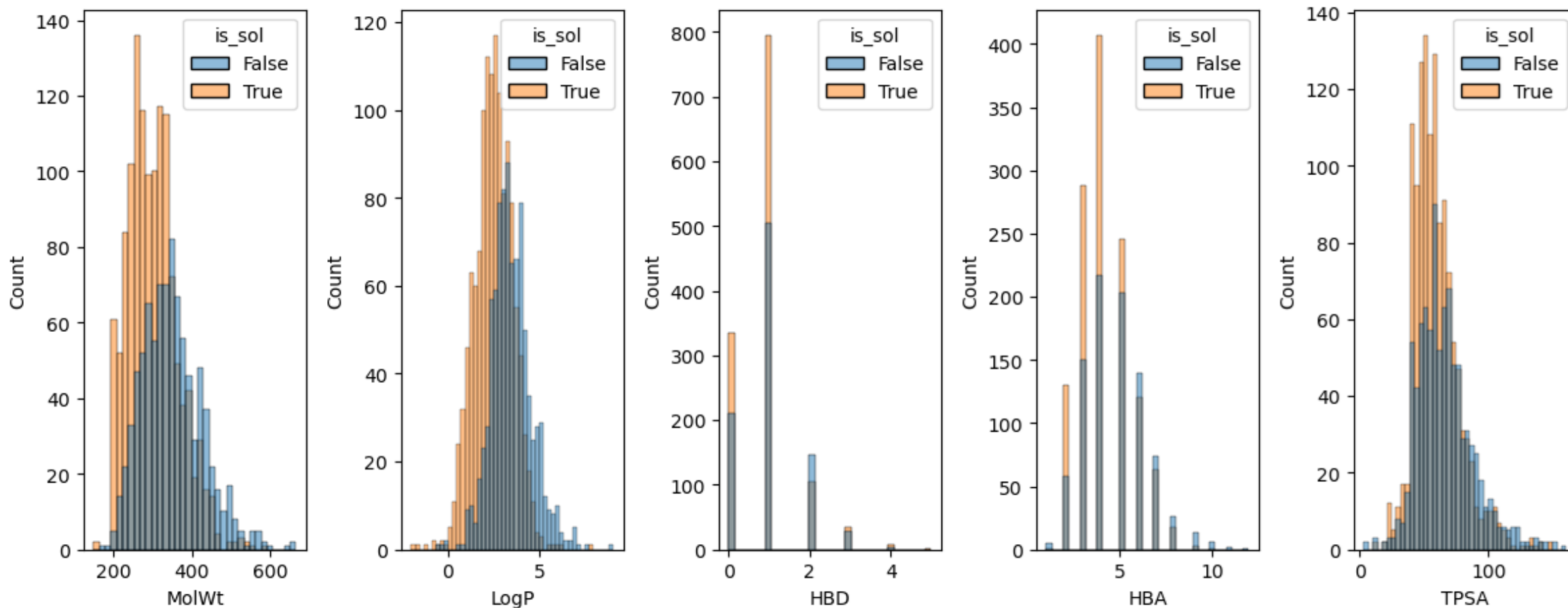
Min = 0.23 μ M

Max = 648 μ M



Fang, Cheng, et al. "Prospective validation of machine learning algorithms for absorption, distribution, metabolism, and excretion prediction: An industrial perspective." *JCIM*, 63.11 (2023): 3263-3274.

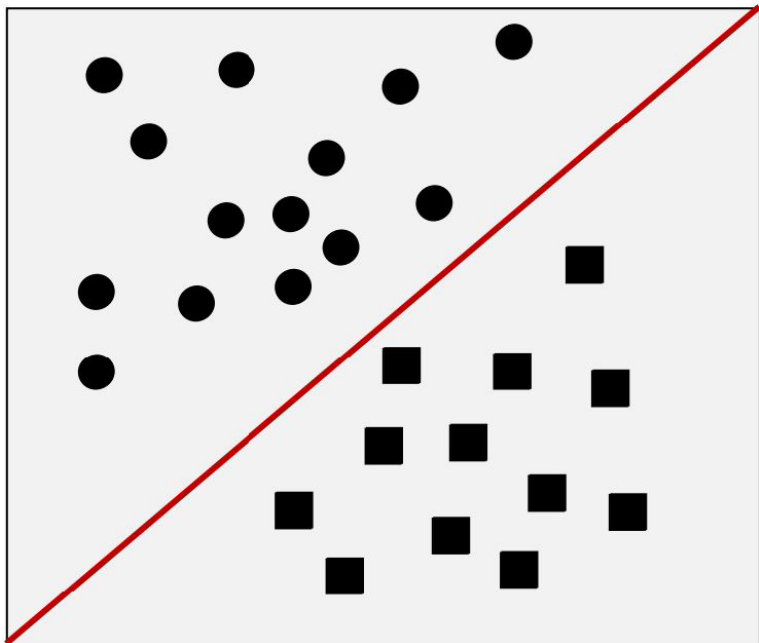
Which Property Best Separates Soluble vs. Insoluble Molecules?



How can ML models help us here?

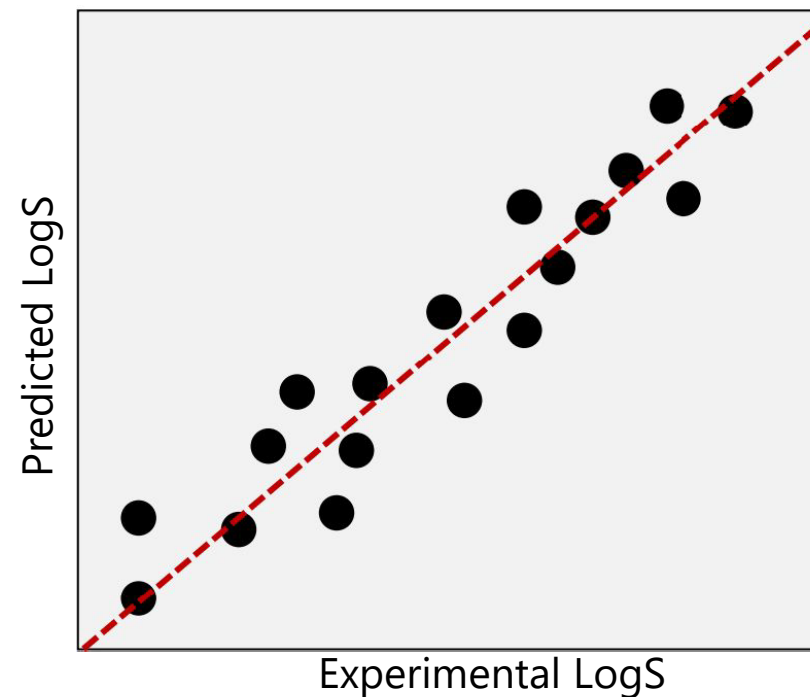
Two Types of ML Models – Classification and Regression

Classification



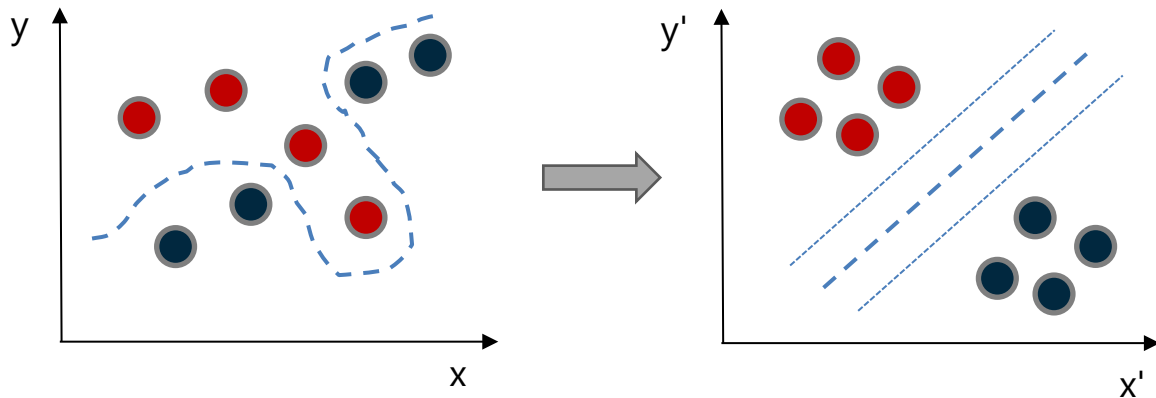
Predict a **category**,
e.g. soluble/insoluble

Regression



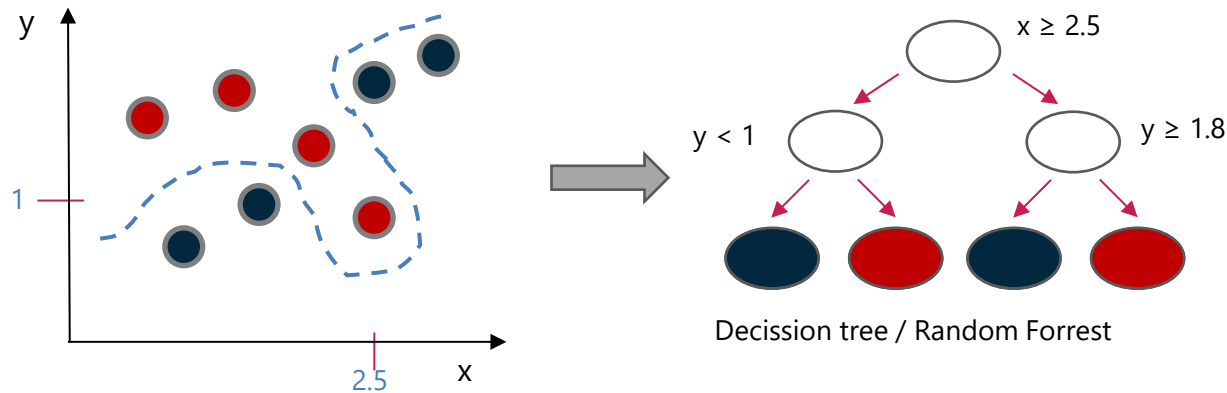
Predict a **value**,
e.g. -4.2

Classification of not linear separable data: Nearest neighbors, **SVM**, **RF**



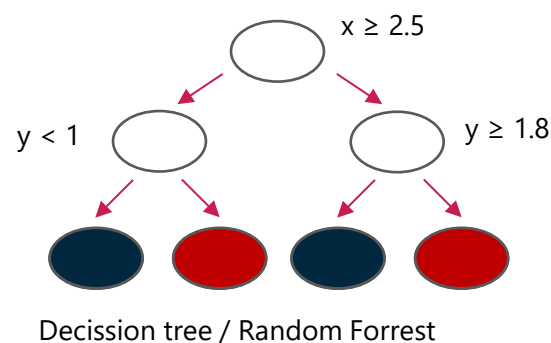
Support Vector Machine (SVM):

- Transformation of original data to higher dimension via Kernel function
- Hyperplane to separate data points



Random forest (RF):

- Combines multiple decision trees
- Random subsets of features for diversity
- Majority voting/averaging



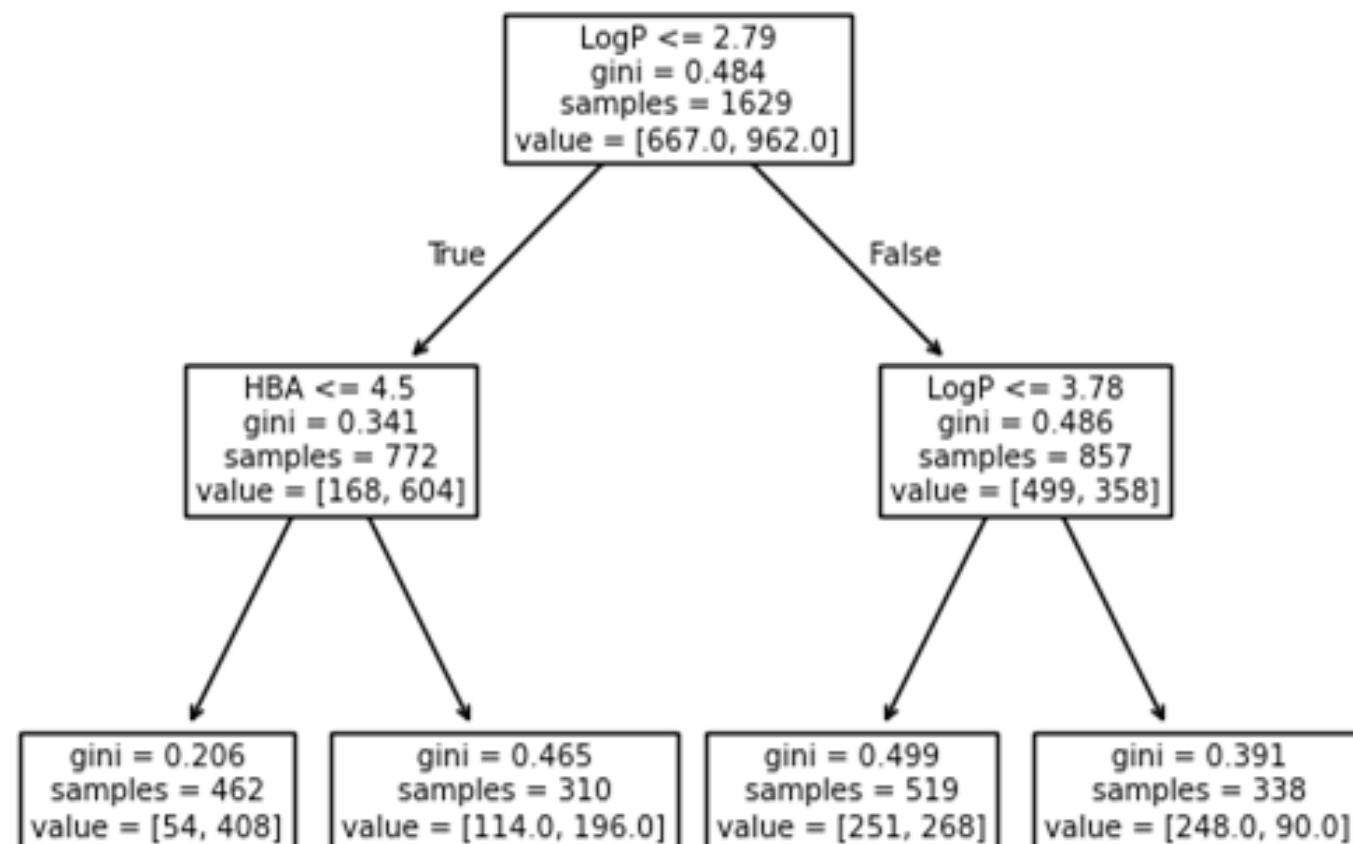
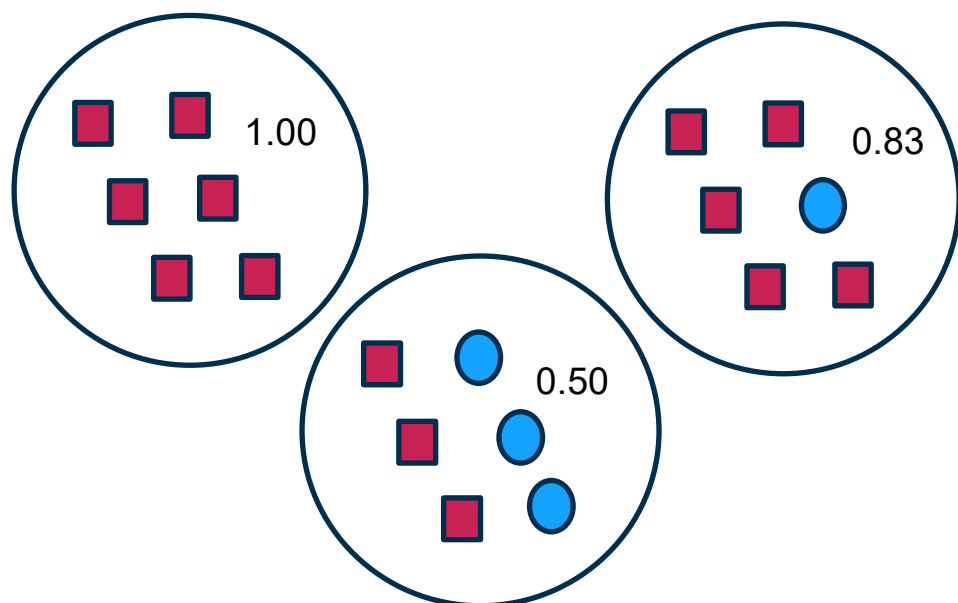
Build a Decision Tree

- Gini Index quantifies the “purity” of a split (*‘how good does this node split the data into 2’*)

$$\text{Gini Index} = 1 - \sum_{i=1}^n (P_i)^2$$

P_i = proportion of samples belonging to class i

n = number of classes



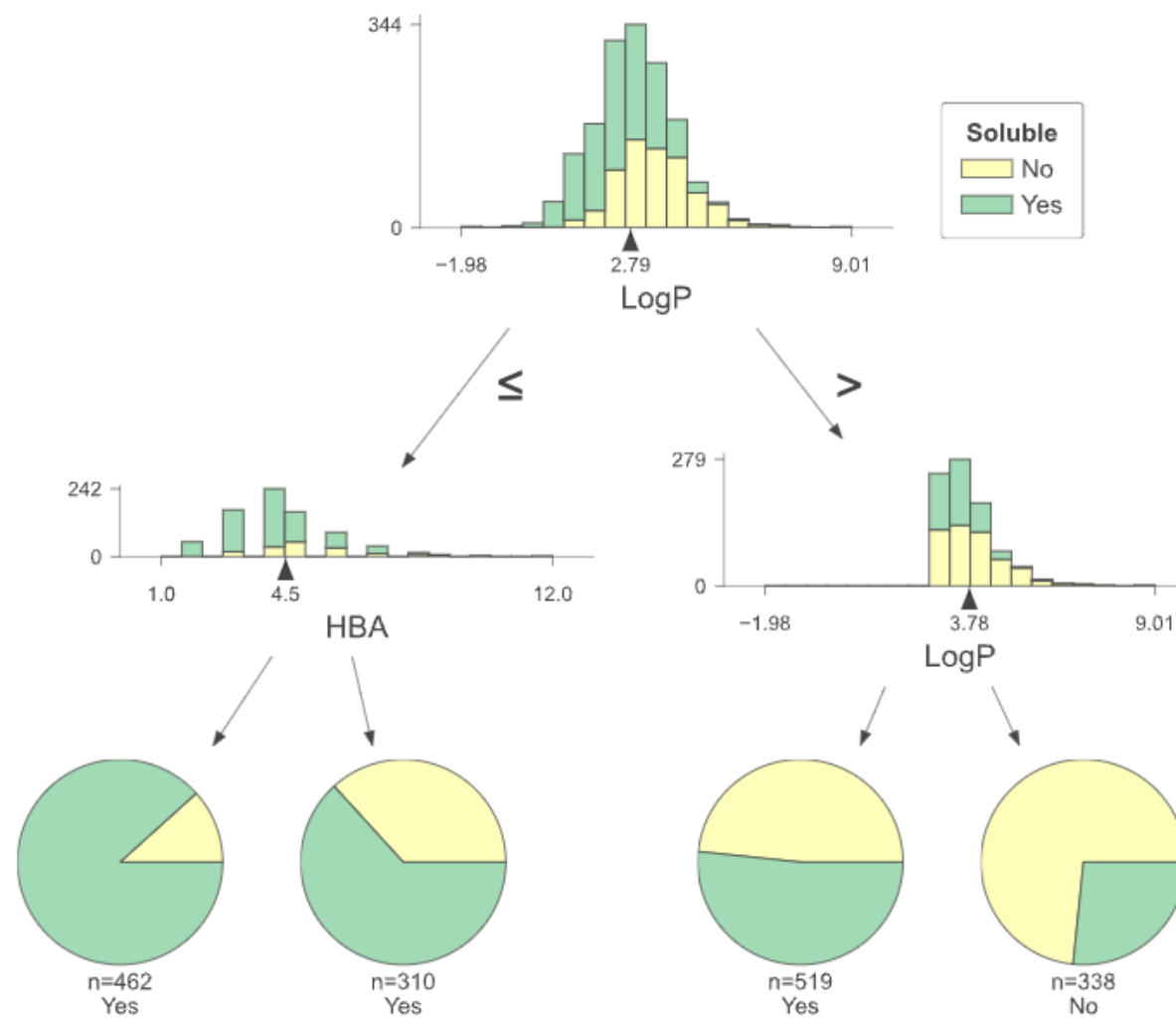
Build a Decision Tree

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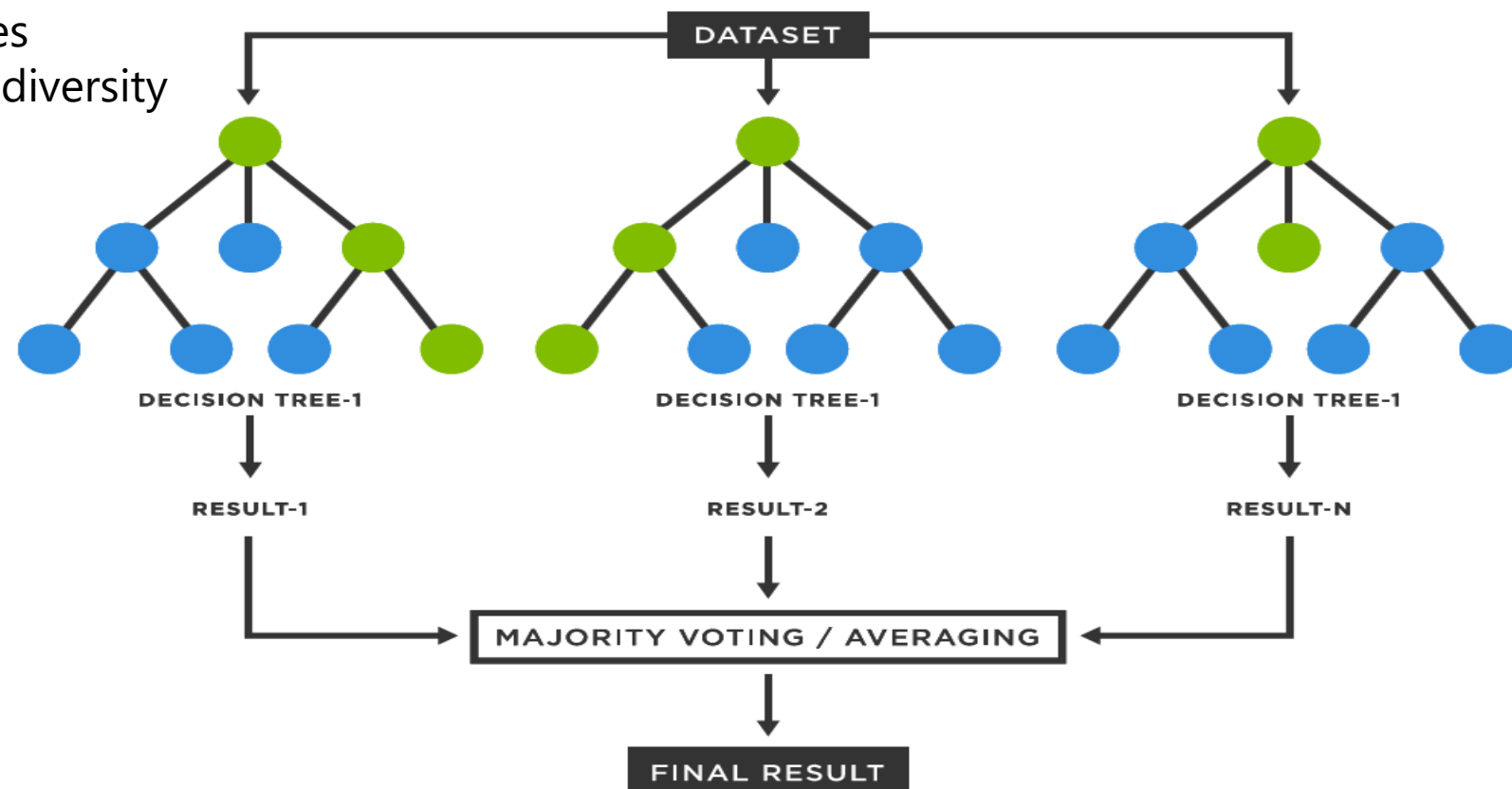
n = number of classes



Random Forest Uses an Ensemble of Decision Trees

Random forest (RF):

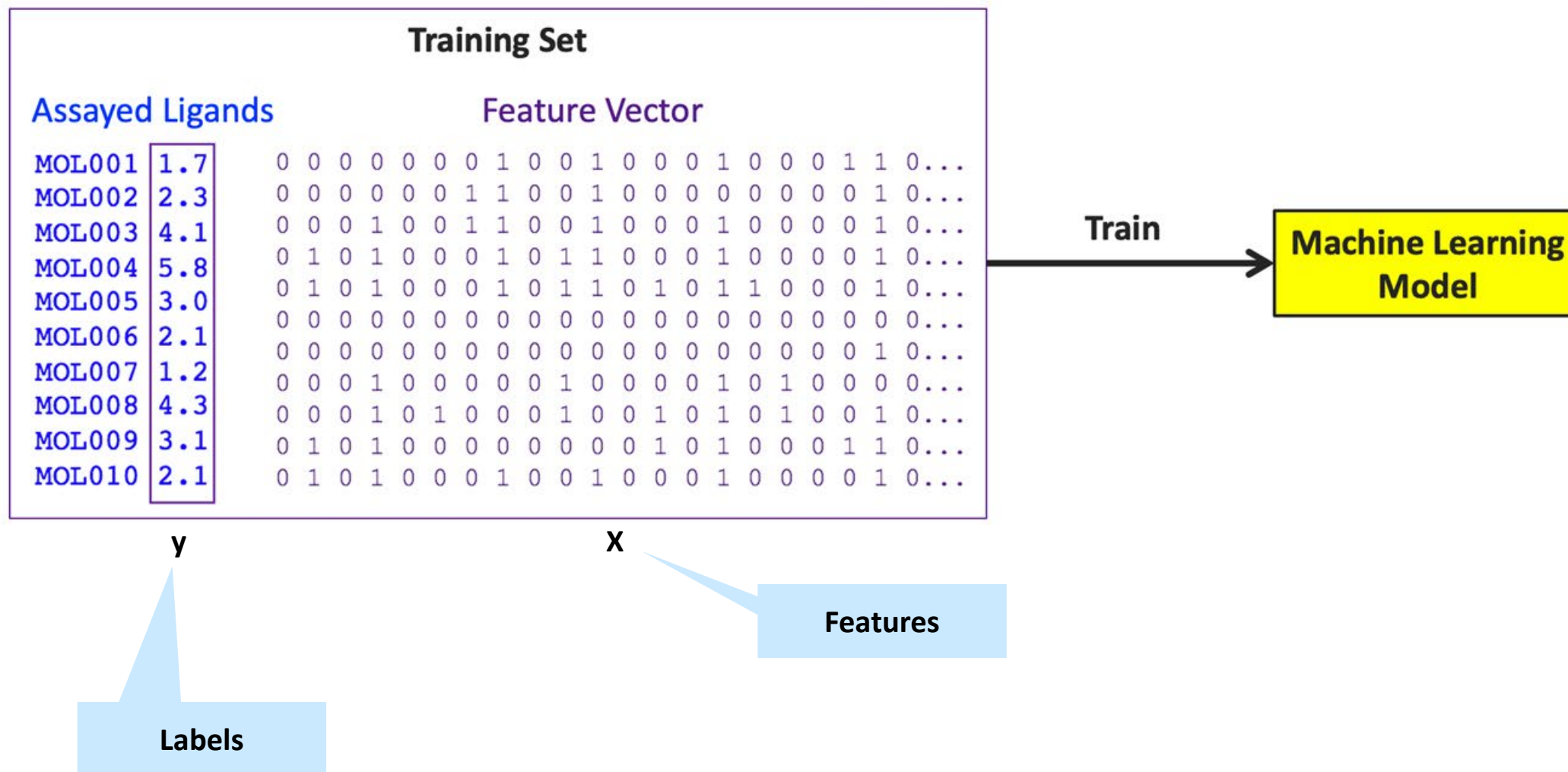
- Combines multiple decision trees
- Random subsets of features for diversity
- Majority voting/averaging



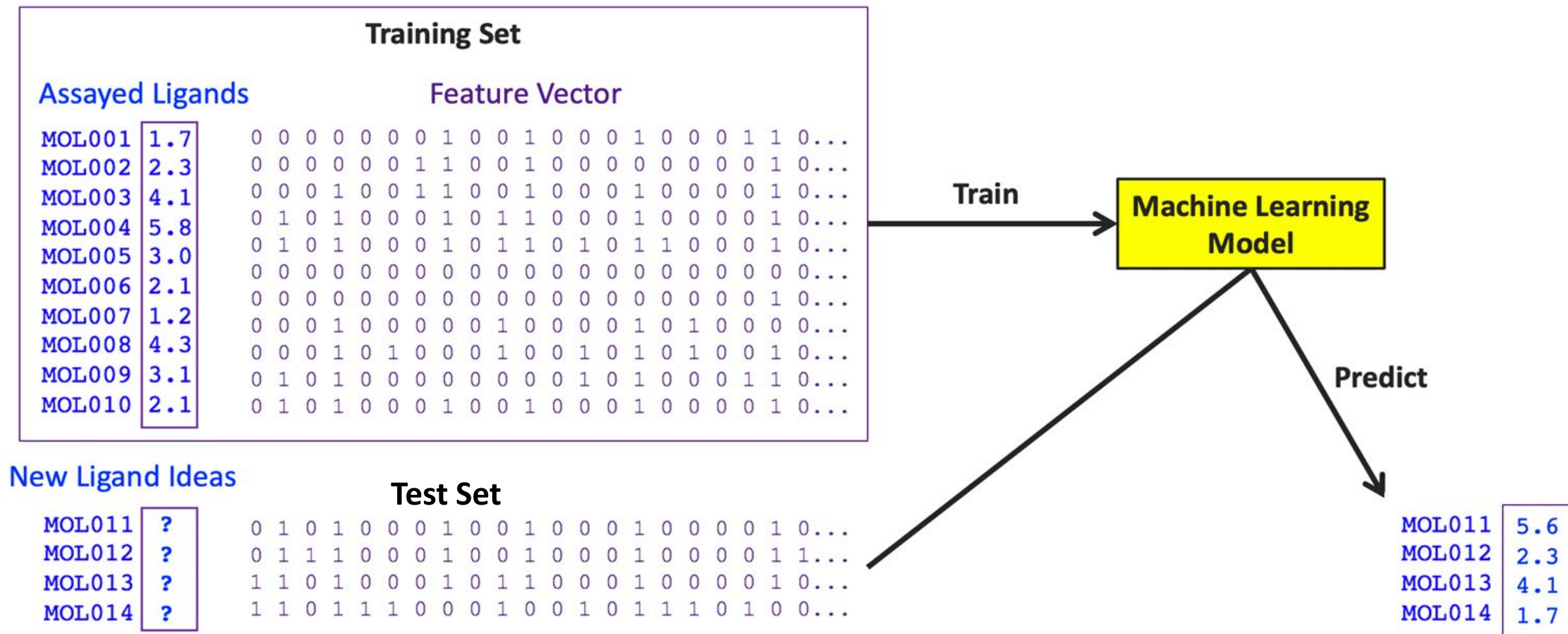
Python library:



Training a Machine Learning Model



Making Predictions With a Machine Learning Model



What we will do in detail ...

Session 1

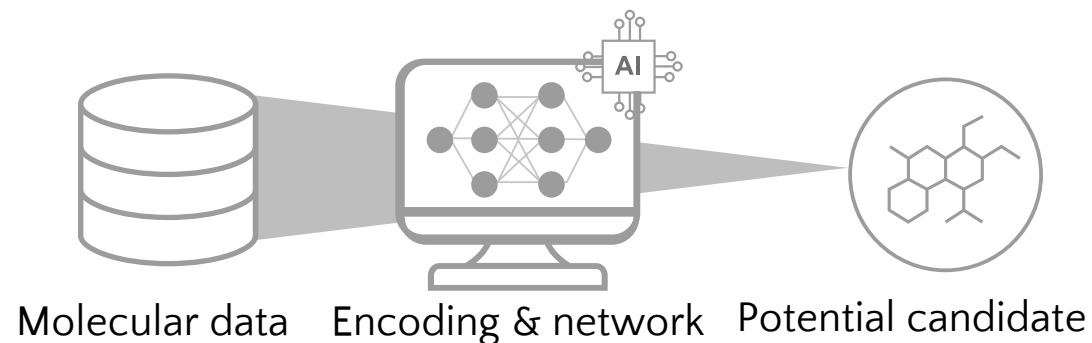
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- **Performance evaluation**

Session 3

- Deep learning
- Neuronal networks
- Molecular encodings and architectures



Training a Model that Generalizes Well to New Data

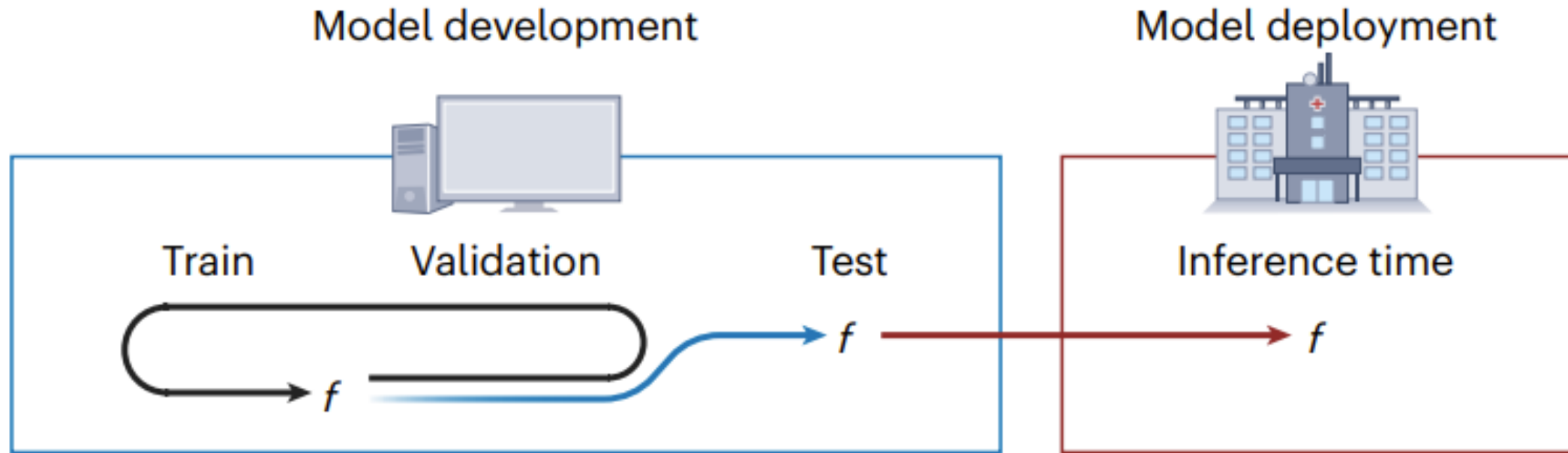


Fig. taken from <https://doi.org/10.1038/s41592-024-02362-y>

Prediction Performance of a **Regression** Model

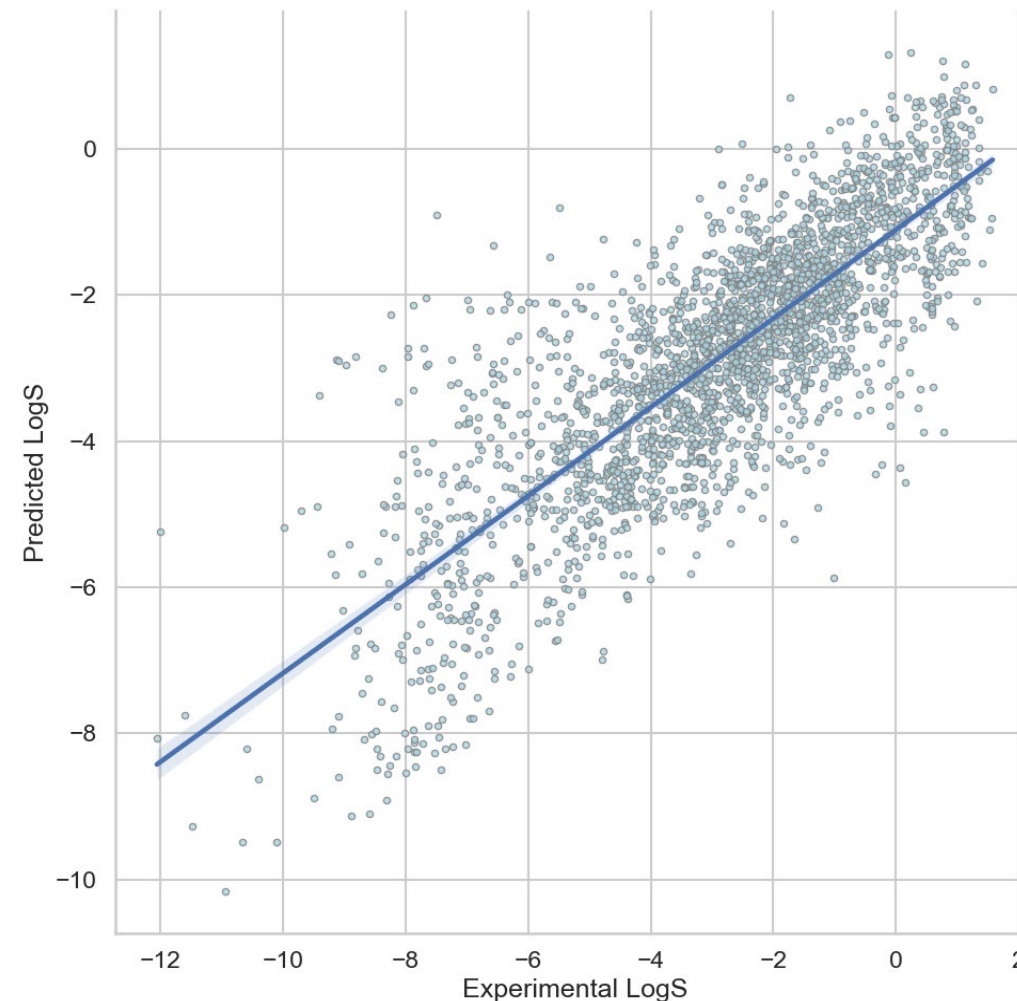
Aqueous Solubility

MSE (root-mean-squared error)

- Average magnitude of the errors between predicted values and actual values
- The lower the better

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- n : Number of data points.
- y_i : Actual value of the target.
- $\hat{y}_i$: Predicted value of the target.



Prediction Performance of a **Regression** Model

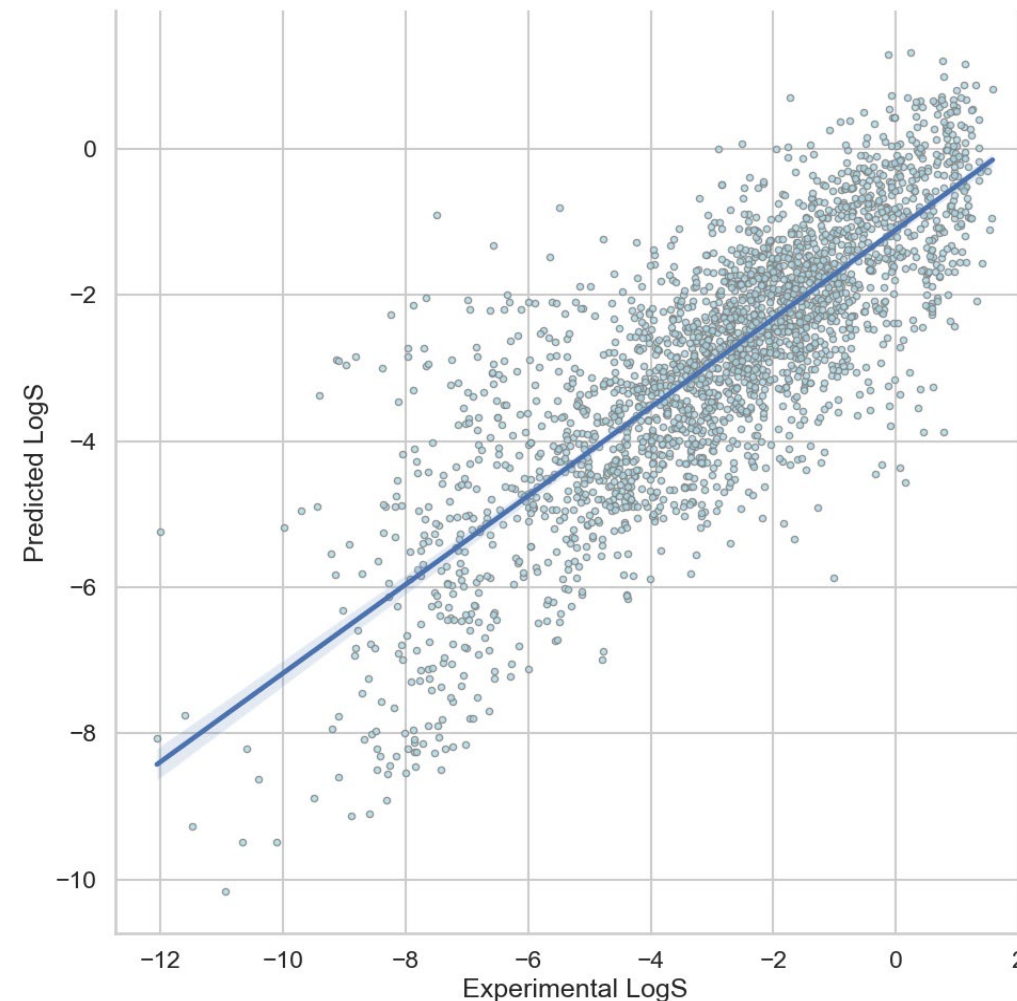
Aqueous Solubility

R^2 (Coefficient of Determination)

- How well does the model explain the variance in the target variable
- Ranges from 0 (no fit) to 1 (perfect fit)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

- y_i = actual value
- $\hat{y}_i$ = predicted value
- $\bar{y}$ = mean of actual values



Prediction Performance of a **Classification** Model

- Performance measures

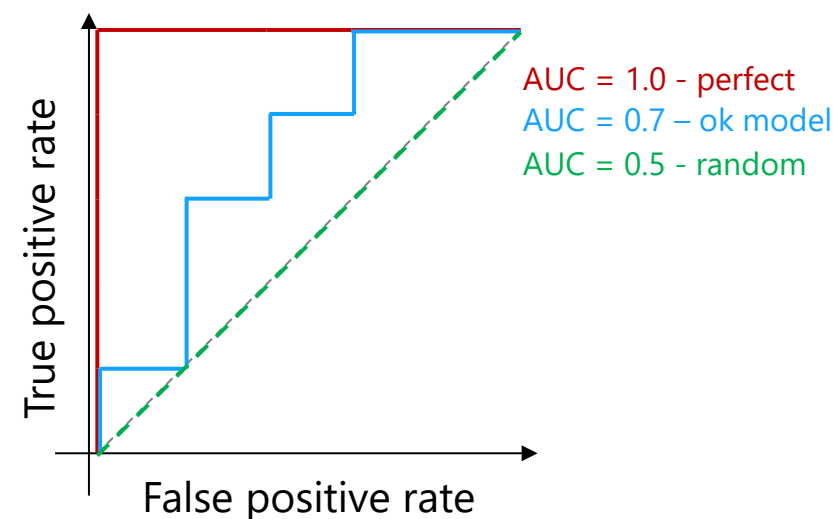
- sensitivity (true positive fraction: TP / P)
- specificity (true negative fraction: TN / N)
- accuracy = $(TP+TN)/(P+N)$

Confusion matrix

Model predicts X	True Active	True Inactive
to be <i>active</i>	True Positive (TP)	False Positive (FP)
to be <i>inactive</i>	False Negative (FN)	True Negative (TN)

- ROC curve (Receiver Operating Characteristic)

- True positive rate is plotted against the false positive rate (rank ordered list)
- AUC (area under the ROC curve): probability that a classifier will rank a positive instance higher than a negative one

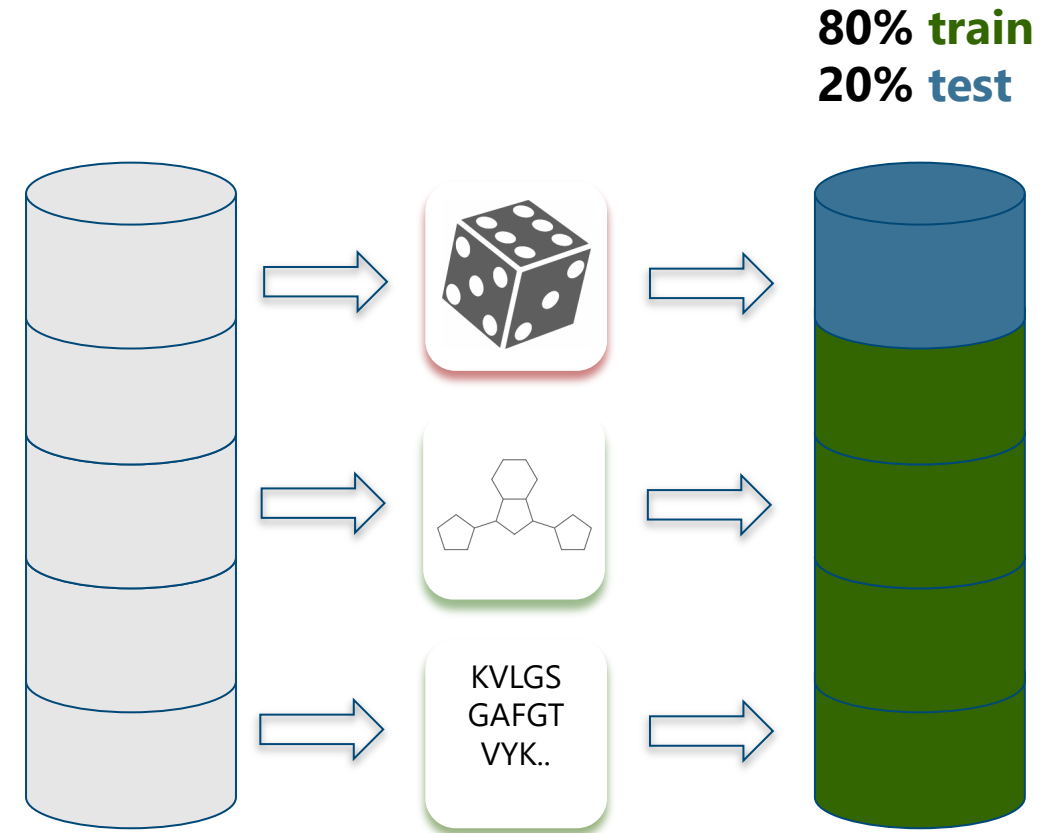


To rate the performance of our ML model,
we need to have **two** sets.

Training and test data: How to split the data?

Data Split is Crucial to Assess Generalizability

- Train/(validate)/test split
- Random splits
 - Often overestimate out-of-domain generalization
- Better: Splits based on
 - Ligand scaffold
 - Assay
 - Protein sequence

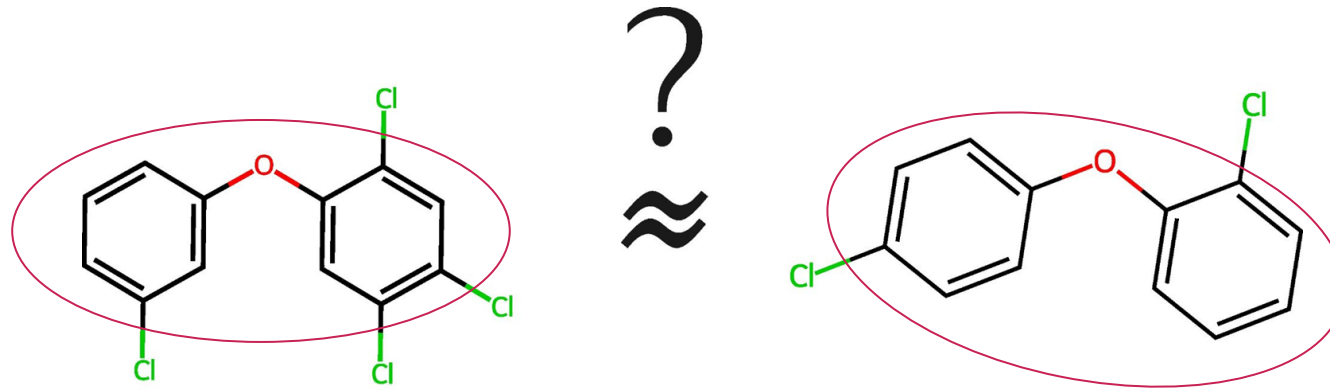


Classification: Consider labels

- Same label ratio in test and training -> Stratified split

Molecular Scaffold - Grouping Molecules

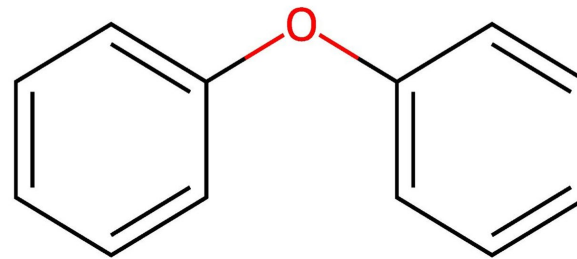
Often molecules are so similar, that inference becomes *too easy*...



Murcko scaffold*:

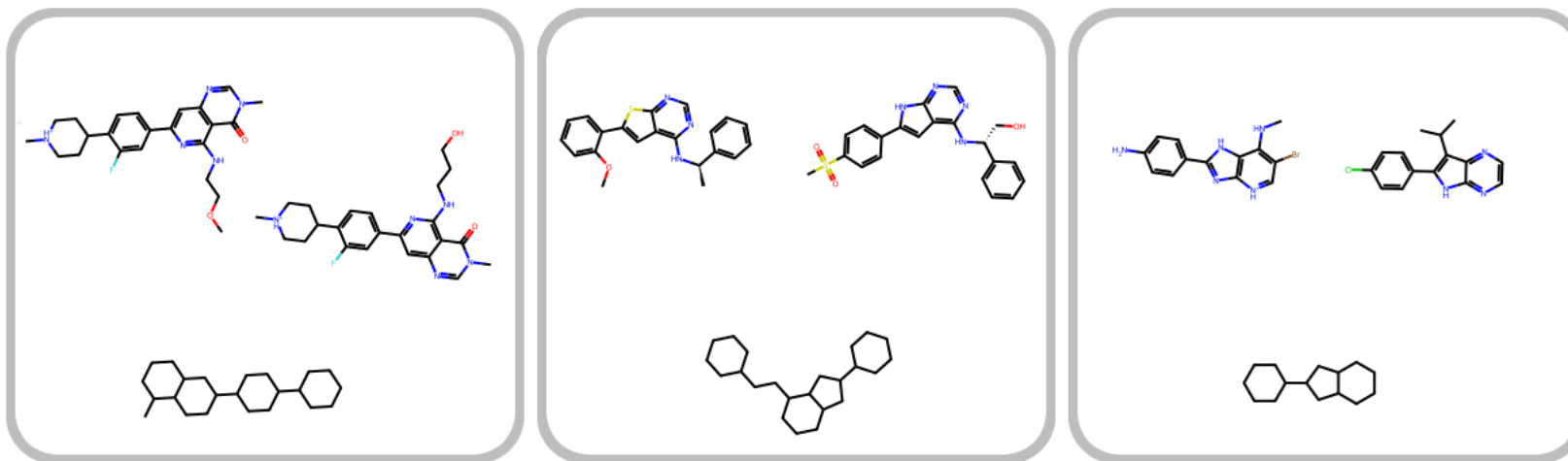
Remove atoms that are not

- In a ring
- Connecting rings



*slight simplification: see [here](#) for details.

Scaffold Split



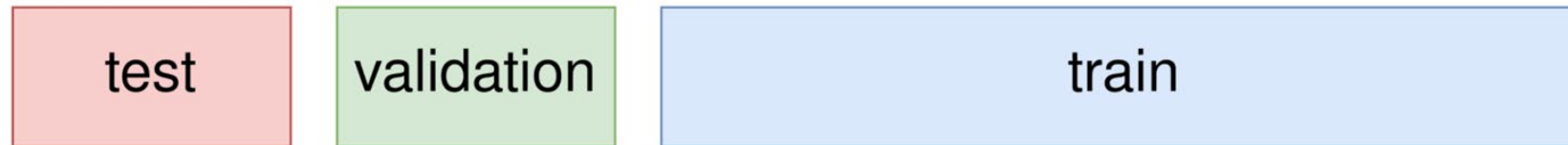
- Avoid data leakage over structurally similar ligands
- Group/partition by scaffold
- Ensure each group only in one of {train, test, val}

5-fold Cross-Validation (with External Test Set)

	dataset					If available
Iteration 1	test	train	train	train	train	ext. test
Iteration 2	train	test	train	train	train	ext. test
Iteration 3	train	train	test	train	train	ext. test
Iteration 4	train	train	train	test	train	ext. test
Iteration 5	train	train	train	train	test	ext. test

Validation Sets - Optimize Hyperparameters **without Cheating**

- Use a validation set alongside training and test set



- **Training:** Fitting the model
- **Validation:** Finding the best model*
- **Test:** Assess generalizability

*When to stop optimizing, which architecture to use, ...

DataSAIL - **Data** Splitting **A**gainst **I**nformation **L**eakage

Developed by Roman Joeres at Prof. Kalinina's lab, Saarland University
uses mathematical optimization to identify the most difficult split

<https://github.com/kalininalab/DataSAIL>



Hands on ...

Time	Topic	Presenter
08:30-09:10	Lecture: From molecular data to models	Andrea Volkamer
09:10-10:00	Hands-on: Data	Katharina Buchthal
10:00-10:40	Lecture: Machine learning (ML) essentials	Andrea Volkamer
10:40-11:30	Hands-on: ML	Katharina Buchthal
11:30-12:30	Break	
12:30-13:00	Hands-on: continued	Katharina Buchthal
13:00-13:40	Lecture: Deep learning (DL) applications	Andrea Volkamer
13:40-14:30	Hands-on: DL	Katharina Buchthal

https://github.com/volkamerlab/cic_summerschool_2025